LODESTAR

DISPATCH ATLAS — FALL 2026

An intelligent freight load board & autonomous dispatch engine

SFI INTERNATIONAL · FALL 2026 ENGINEERING MINI-CHALLENGE

A SUBMISSION BY

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LIVE DEMO

lodestar-production-4820.up.railway.app

A CAPITAL-ALLOCATION DECISION EVERY FEW MINUTES, ON A TEN-MINUTE CLOCK

This truck, at this hub, for the next two days, against this payout. Three things make it hard, and none of them is “find a truck”.

THE COST OF A LOAD IS NOT ON THE LOAD

The posted estimate covers loaded miles only. The empty move depends on which truck goes, and where that truck will be, not where it is now.

CAPACITY IS A SCHEDULE, NOT A COUNT

“Seven units at Atlanta” says nothing about reaching a 06:00 pickup 90 miles away and still making the Friday load one of them already promised.

EVERY BOOKING MOVES THE NETWORK

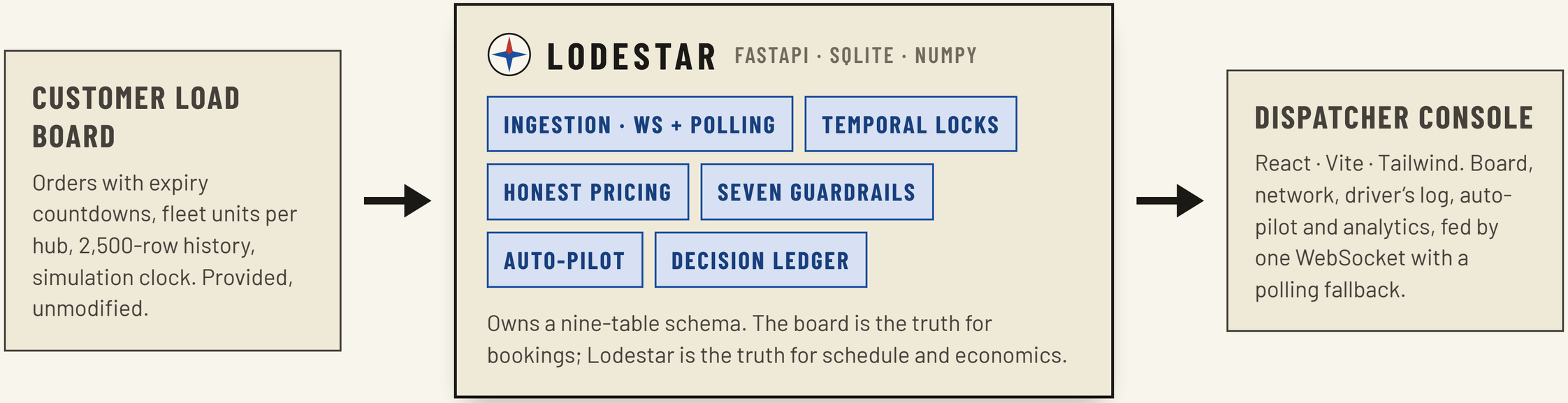
Capacity follows the freight: the truck that takes ATL → CHI becomes Chicago capacity, and Atlanta is one unit thinner.

WHAT THE BOARD CHECKS, AND WHAT LODESTAR REFUSES

CONCERN	THE BOARD ENFORCES	LODESTAR ADDS
Weight	44,000 lb max, HTTP 409 beyond it	Caught before the call; never overridable
Capacity	A free unit exists at the hub	A unit that can reach the pickup on time and is free for the whole window
Empty miles	Nothing	Deadhead measured from the unit the board will bind, at its projected drop; capped by policy
Margin	Nothing	Floor in dollars and percent, after loaded and empty cost
Reserves	Nothing	Reserve units per hub and a lock-aware 24 h forecast
Bids	Random award	EV-optimal bid above a cost-plus floor, with P(win)
Audit	Nothing	A ledger entry for every accept, bid, decline and skip

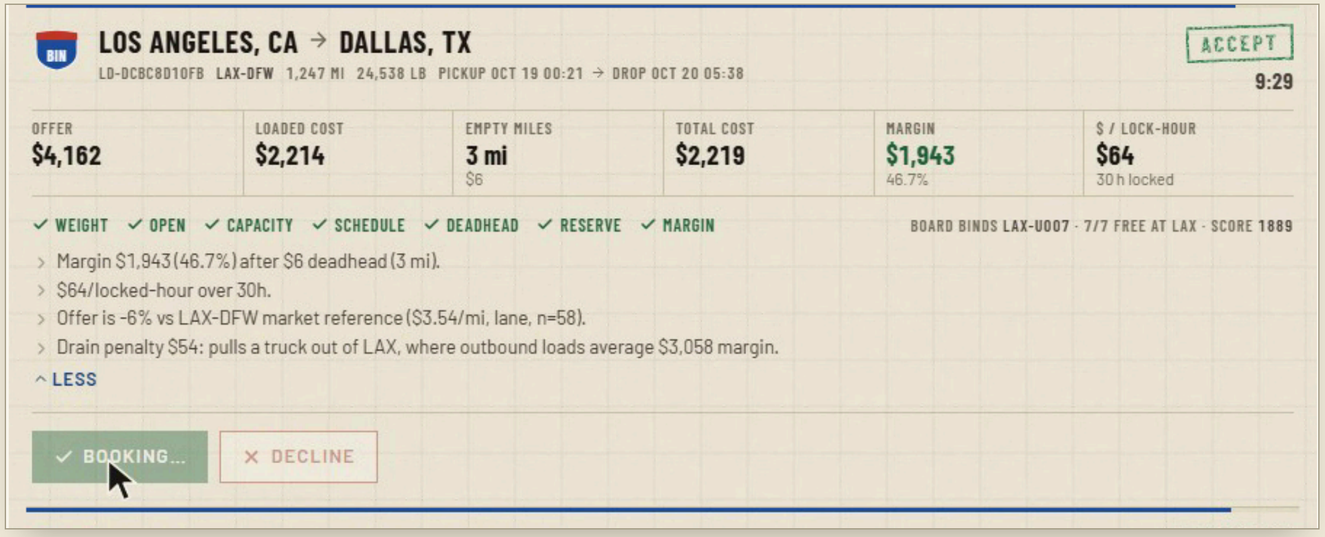
Everything between **“the board will let you”** and **“this is good business”** is where Lodestar lives.

LODESTAR SITS BETWEEN THE CUSTOMER BOARD AND THE FLEET



Drawn as a motor carrier's road atlas: map paper, ink rules, road-hierarchy colour. One Docker image runs both services, live on Railway.

EVERY CARD CARRIES THE WHOLE DECISION BEFORE YOU CLICK



A fixed-rate card after Accept: the board's own confirmation lands as an inked stamp.

HONEST ECONOMICS

Offer, loaded cost, empty miles from the unit the board will actually bind, total cost, margin in $ and %, and margin per locked hour so a two-day load competes fairly with a six-hour one.

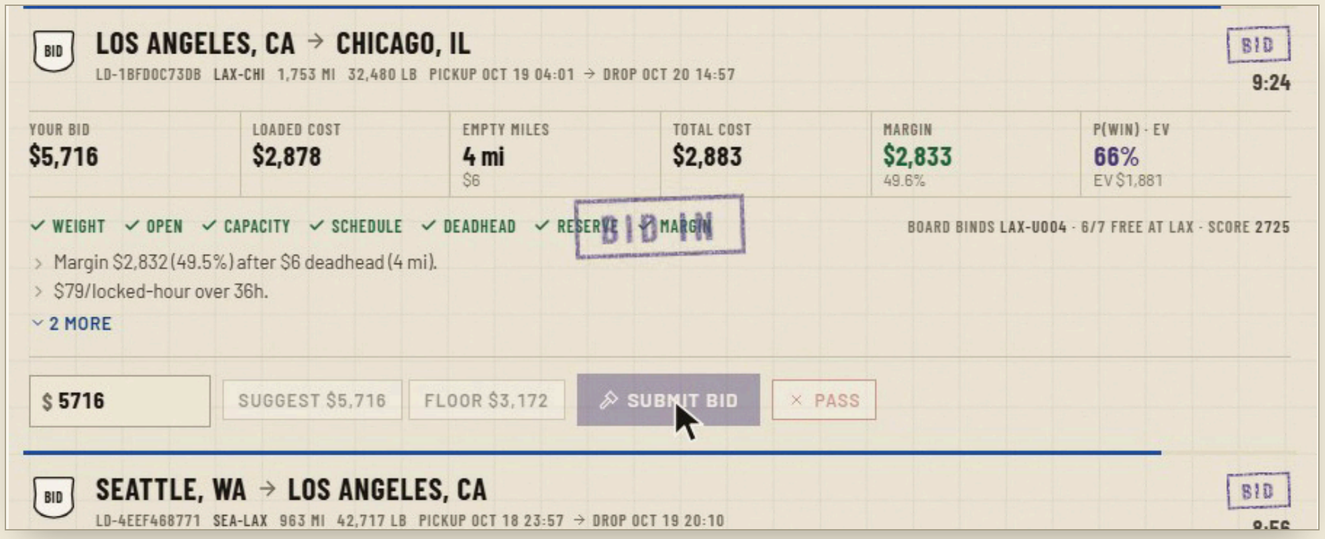
SEVEN GUARDRAILS

Weight · open · capacity · schedule · deadhead · reserve · margin, with the engine's reasons in a dispatcher's own words.

REFUSED BEFORE THE BOARD IS CALLED

A dispatcher can override the business rails with a note. Overweight and expired loads cannot be overridden, and auto-pilot never can.

BID TO PROTECT MARGIN WHEN WE WIN, NOT TO WIN EVERY AUCTION



The bid input is prefilled with the model's suggestion; P(win) and EV update live as you edit it.

HELD-OUT AUC

0.67

VS HISTORICAL BIDS

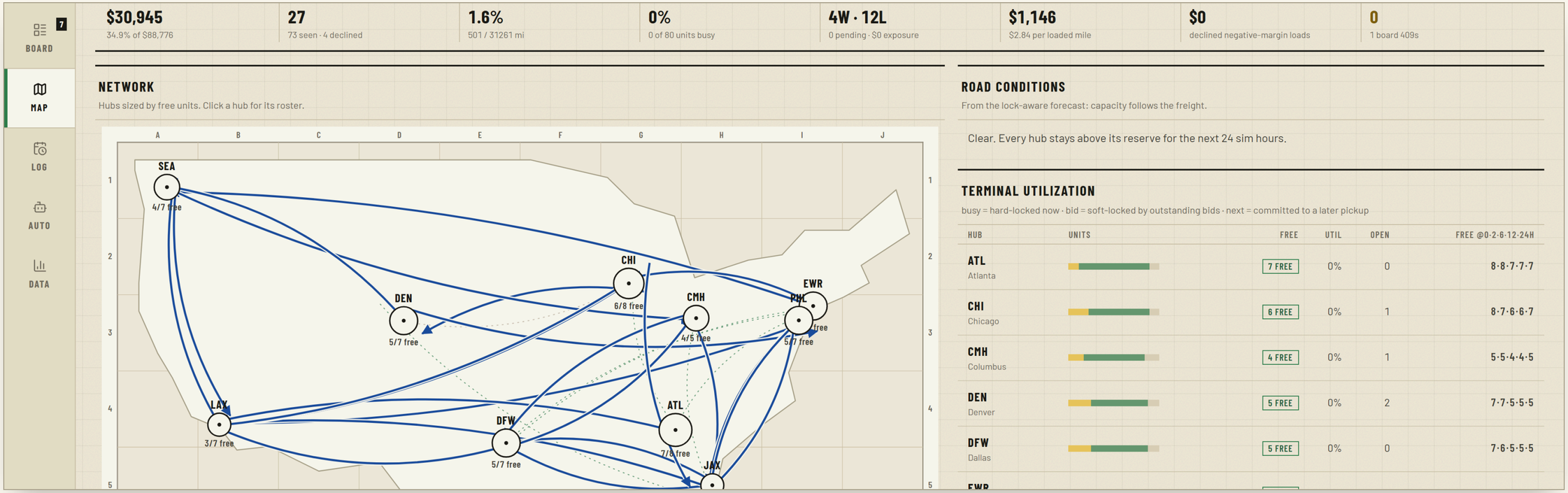
+16%

A numpy logistic regression on 1,081 historical auctions: bid ÷ market − 1, log miles, weight and season. Brier 0.23, calibrated across buckets; the bid-gap coefficient is strongly negative, as it should be.

The suggested bid maximises **P(win) × (bid − total cost)** above a cost-plus floor. On the held-out 25 % it yields $714 expected profit per auction against $616 for the historical bids, never below cost.

Lane market rates and hub outbound value come from the same history and feed the chaining score.

CAPACITY FOLLOWS THE FREIGHT

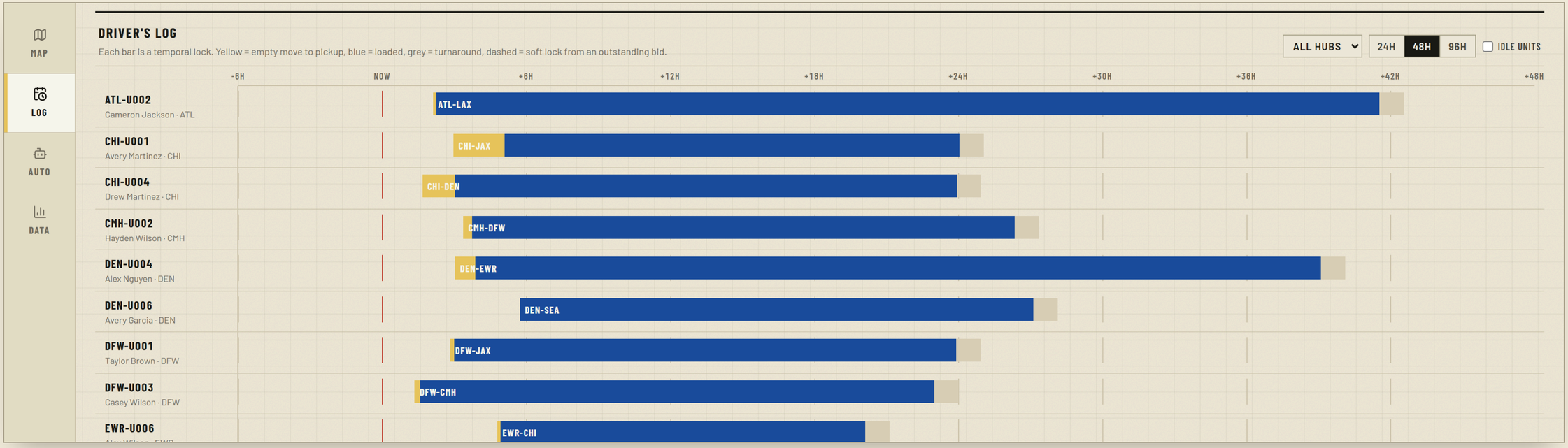


Hubs are sized by free units and coloured by reserve status; click one for its roster.

Blue arcs are booked lanes with the unit dot moving along them; **dashed purple** is an outstanding bid.

Road conditions come from a lock-aware forecast that counts an inbound truck at its destination, so a hub is flagged before it runs dry.

ONE TEMPORAL LOCK PER BOOKING, AND DOUBLE-BOOKING CANNOT HAPPEN

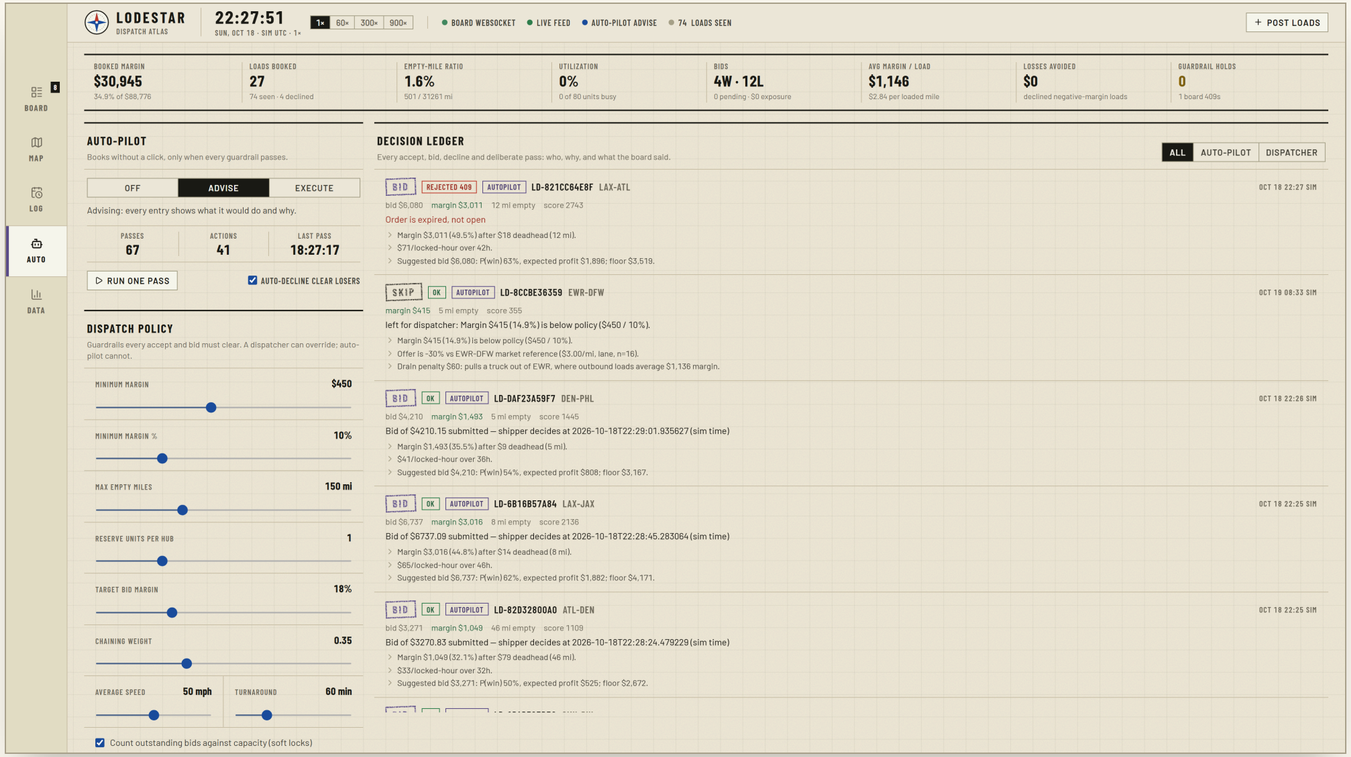


A lock is **[pickup – deadhead travel, delivery + turnaround)**: hard when the board confirms a booking, soft (dashed) while a bid is outstanding.

Feasibility is computed from the unit's **projected position** at pickup time: the last lock ending before it says where the truck will be and when it is free.

So the next load is planned from the previous drop, chaining is structural, and an overlapping window is refused before the board is called.

HELD TO THE SAME GUARDRAILS AS A HUMAN, AND IT SHOWS ITS WORK



OFF · ADVISE · EXECUTE

Scores the open board every two seconds and books best-score-first, not first-to-expire.

PREFERS PROFITABLE UTILIZATION

Clear losers are declined, borderline loads are left on the board for a person.

EVERY ACTION AND NON-ACTION IS LOGGED

Accept, bid, decline and skip entries carry the same reasons a dispatcher sees. Policy is live: raise the minimum margin and every open card re-grades.

A SCHEMA IT OWNS, A PURE EVALUATOR, AND PROOF IT WORKS

RELATIONAL SCHEMA

Nine tables: shipments, fleet units, **unit locks** (the temporal ledger), decisions, inventory snapshots, policy. Board unit ids are reused so sandbox and product state never diverge.

REAL-TIME INGESTION

Board WebSocket for instant orders, 3 s reconciliation polling, 4 s fleet sync. Survives socket loss, board restarts and re-seeds; the sim clock is mirrored and re-synced every poll.

CHAINING-AWARE SCORE

score = margin + chain_value − drain_penalty

Rewards delivering a truck into a hub that is short and pays well outbound; penalises draining a hub that is itself valuable. Recommendation is categorical: accept, bid, hold, decline.

PROOF

18 unit tests (locks, evaluator, pricing, sync) · **23**-check end-to-end suite against both servers · a policy backtest over the history · CI that runs tests, builds the console, builds the image and boots it · one image live on Railway.

ONE SESSION AGAINST THE LIVE BOARD, DEFAULT POLICY

LOADS BOOKED

11

of 265 ingested, every one within one poll

BOOKED MARGIN

$20.5k

35 % of revenue

EMPTY-MILE RATIO

1.8%

deadhead ÷ loaded miles

BOARD REJECTIONS

0

every bad accept or bid blocked before the call

Honest deadhead moved the margin figure by an average of 28 empty miles (≈ $50) per load, and by over 100 miles on off-hub “Near …” pickups.

Of ~120 evaluated fixed-rate loads, 3 % were loss-making after deadhead and 11 % fell under the policy floor; the first group is declined automatically, the second left for a human.

Ingest lag ≤ 3 s on the polling path, under 100 ms over the socket. All 80 units’ payout and re-homed hub matched the board after delivery.

THE SEAMS ARE ALREADY CUT WHERE THE SERVICES WOULD SPLIT

CONCERN	TODAY	ENTERPRISE
Storage	SQLite (WAL)	Postgres; a GiST exclusion constraint on (unit_id, tstzrange) makes double-booking impossible at the database, not just in code
Services	One process: ingest, engine, auto-pilot, API	A stream consumer per board, stateless engine workers, a leader-elected auto-pilot, the push hub on Redis pub/sub
Concurrency	Single asyncio loop	Optimistic locking on shipments and SELECT … FOR UPDATE on candidate units, so two dispatchers cannot book the same truck
Optimisation	Greedy best-score-first with projected positions	Periodic global re-plan (ILP / column generation over the next 48 h) behind the real-time engine; the evaluator is already pure and per-hub parallel
Pricing	Logistic regression retrained at boot	Gradient-boosted model retrained nightly on realised outcomes, lane-level drift monitoring, a human-in-the-loop floor
Positions	Interpolated from locks	ELD / telematics feed replaces interpolation; hours-of-service rules become one more guardrail

FOUR KNOWN SEAMS, EACH WITH A NAMED NEXT STEP

THE SANDBOX BINDS THE UNIT

A production TMS would own assignment, and the “best unit” Lodestar computes would be the one dispatched. The code already computes both and shows the difference on the card.

TRAVEL TIME IS DISTANCE ÷ SPEED

A fixed turnaround, no hours-of-service, no traffic. A telematics feed and an HOS guardrail are the planned replacements.

THE MODEL LEARNS FROM SYNTHETIC HISTORY

Lane medians stand in for a market reference, which caps the win model near 0.67 AUC. Real tender data would be richer, and the training path is the same.

CHAINING IS ONE STEP AHEAD

The score looks one hop into the network; it is not a global plan. The periodic re-plan on the previous slide is where that goes.



LODESTAR

DISPATCH ATLAS

EVERY DISPATCHER, THE BEST ONE ON THE DESK, ON THE WORST DAY

LIVE DEMO

lodestar-production-4820.up.railway.app

SOURCE · DOCS · REPORT

github.com/subodh-thallada/lodestar-dispatch

SUBODH THALLADA

SFI International · Fall 2026 Engineering Mini-Challenge